

MSE 311 Thermodynamics of materials-Quiz 7

Name:

Student ID:

1. Please explain the difference and relationship between *Raoult's law* and *Henry's law* in short words, respectively.

Raoult's law: $p_A = X_A p_A^\circ$ *Henry's law:* $p_A = k^A X_A$

The range of composition over which the solute B obeys Henry's law, the solvent A obeys Raoult's law

2. Choose the right selection.

If the solution is positive deviation from Raoultian behavior, then

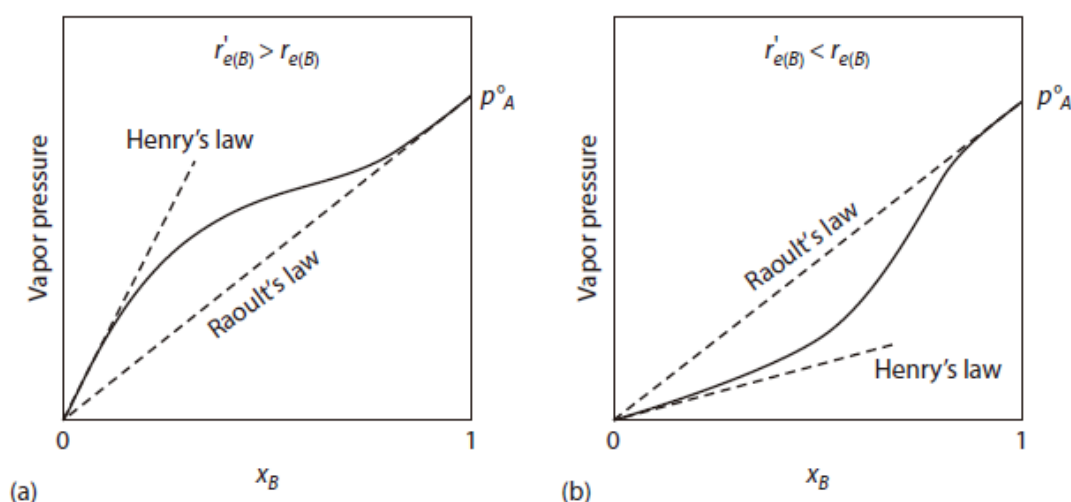
a. γ_i (☒ $>$ ☐ $=$ ☐ $<$) 1,

b. $\Delta \bar{H}_i^M$ (☒ $>$ ☐ $=$ ☐ $<$) 0,

c. A-B bonding is (☐ stronger ☒ weaker) than A-A,

d. The mixing is (☒ endothermic ☐ exothermic),

e. the change of the vapor pressure in (☒ Fig. a ☐ Fig. b) is right.



3. Write down the change of enthalpy under the *regular solution* and the change of enthalpy under the *statistics model of solutions*.

Regular solution: $\Delta H^M = \Omega_{AB} X_A X_B$

Statistics model of solution: $\Delta H^M = zN_0 X_A X_B \left[E_{AB} - \frac{1}{2}(E_{AA} + E_{BB}) \right]$